Dane Pearson

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Education

University of California, Santa Cruz **September 2022 – August 2026**
B.S. in Cognitive Science, Minors in Statistics and Computer Science | GPA: 3.76

- **Honors:** Dean's Honors (6 quarters)
- **Coursework:** Real Analysis, Probability Theory, Classical and Bayesian Inference, Statistical Data Analysis (graduate), Linear Algebra, Calculus I–III, Machine Learning, Deep Learning, Data Structures and Algorithms, Database Management

Research Experience

Department of Statistics, UC Santa Cruz **February 2026 – Present**
Undergraduate Researcher | Advisor: Prof. Sangwon Hyun

- Built the data foundation for a study of how ocean conditions drive phytoplankton populations, designing the end-to-end workflow that produces the analysis-ready covariate data PhD students and postdocs use to fit a sparse mixture-of-regressions model (flowmix; Hyun et al., 2023).
- Designed an adaptive matching procedure to align environmental measurements with research-cruise tracks across mismatched sampling times and locations, ensuring each sample's covariates reflect the conditions it actually experienced; evaluated per-variable coverage to decide which of 12,800 candidate variables to retain.
- Created a reproducible covariate data system for the lab, enabling members to regenerate analysis-ready datasets as new expeditions are added; scaled it across 88 research expeditions with parallel cluster jobs, documented the workflow, and presented it at an hour-long lab-wide meeting.
- Built automated integrity checks that caught mismatched and duplicate records before they reached collaborators' analyses, including a case where the source database silently returned the wrong dataset; diagnosed deprecated and broken source tables and reported them to Prof. Hyun.
- Helped screen cytogram time points for censoring and instrument errors, using time-series heatmaps and two-dimensional scatterplots to identify points to exclude before model fitting.

Presentations

Hyun Lab Meeting, Department of Statistics, UC Santa Cruz **March 2026**
"Building a Common Environmental Covariate Set for Cross-Cruise Phytoplankton Mixture Modeling"

Projects

COVID-19 Mortality Analysis in R **December 2025 – November 2026**
STAT 204 (graduate) team project, independently extended | Repository

- Investigated how age, sex, region, and year shape U.S. COVID-19 mortality rates (2020–2023), building person-year denominators in R by joining 138,000+ rows of CDC death counts with age- and sex-specific Census estimates.
- Built models iteratively from diagnostics: a dispersion test rejected Poisson in favor of negative binomial, and DHARMa residuals revealed state-level clustering, resolved with a glmmTMB mixed model (state random intercepts).
- Found that CDC suppression of counts 1–9 biased child mortality estimates; multiply imputed 1,136 suppressed counts and pooled with Rubin's rules, confirming that adult findings were robust.
- Estimated adjusted mortality rate ratios with 95% CIs: 8.8× for ages 85+ vs. 65–74, 53% higher for males, and 26% higher in the South than the Midwest; confirmed an age-by-sex interaction by likelihood ratio test.

Professional Experience

Siemens **May 2026 – August 2026**
Database Operations Intern

- Proposed and built an automated Python pipeline consolidating 3,500+ AWS server and database records into a self-updating dashboard; worked with cloud engineering teams to define how environments should be categorized, then designed rule-based classification that reduced records with unknown environments from 2,300+ to under 20.
- Developed a Python web app for a 12-person database team that summarized ticket data and used internal Claude to generate automated root-cause analysis and assignment recommendations; translated loosely defined team needs into specific metrics through feedback rounds, and presented it to managers and engineers at cloud operations meetings.
- Led 4 knowledge-transfer sessions to hand off 2 projects to global teams, explaining system design and maintenance to engineers and non-technical stakeholders so the teams could run both tools independently after my internship.

Skills

Statistics:	Generalized linear and mixed-effects models (Poisson, negative binomial, random effects, offsets, overdispersion, rate ratios), linear regression, model diagnostics, interactions, likelihood and Bayesian inference, model selection (AIC, likelihood ratio tests), multiple imputation, PCA
Languages:	R, Python, SQL, MATLAB, LaTeX
Libraries:	tidyverse (dplyr, tidyr, ggplot2), MASS, glmmTMB, DHARMa, broom, pandas, NumPy, plotly, PyTorch
Computing:	HPC (Slurm), Linux, Git/GitHub, Quarto, PostgreSQL

References

Prof. Sangwon Hyun sangwonh@ucsc.edu
Assistant Professor, Department of Statistics, UC Santa Cruz

Dr. Robert Hingtgen rhingtge@ucsc.edu
Lecturer, Department of Mathematics, UC Santa Cruz

Prof. Marcela Alfaro-Cordoba macordob@ucsc.edu
Associate Professor, Department of Statistics, UC Santa Cruz